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Paper Code : PEC-CS702B/PEC-IT702C/PECDS702B Soft Computing

UPID : 007556

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is perceptron?

(II) What is the name of the process that represents modified elements of the DNA?

(III) Define Hard computing.

(IV) _____ represents the fuzzy logic.

(V) What do you mean by NN architecture?

(VI) The entire combination of genes is called_____.

(VII) How can uncertainty be represented?

(VIII) What is cardinality of a Fuzzy set?

(IX) What is auto associative memory?

(X) Matrix crossover is also known as One dimensional. State True/False.

(XI) An approximation based fast solution system can implemented using Fuzzy logic . State True/False.

(XII) What is mean by Fuzzy logic controller?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Define Fuzzy set and write basic fuzzy set operations. [5]
3. Discuss linear separable problem in ANN. [5]
4. Explain mutation operator & its significance. [5]
5. What is soft computing? What are the different application areas of soft computing? [5]
6. Compare classification and clustering. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Find the Fuzzy Hamming distance and Euclidean distance between given two fuzzy set: A = { (x1, 0.4), (x2, 0.8), (x3, 1.0), (x4, 0.0)}, B = { (x1, 0.4), (x2, 0.3), (x3, 0.0), (x4, 0.0) }. [7]

   (b) Suppose we have 2 universes for a thermistor problem and it is described by a collection of elements: X = {1, 2, 3}, Y = {1, 2, 3, 4, 5}. A crisp set A is defined on universe X and a crisp set B on universe Y as follows: A = {1, 3}, B = {2, 4, 5}. Find the values of R where the inference given as 'If A then B: R'. [8]

8. (a) Define Tautology. Prove the tautology (P=>Q) V (Q=>P) using truth values in fuzzy logic. [4]

   (b) Describe Sugeno Fuzzy Inference System (FIS) with an examples. [4]

   (c) An athletic race was conducted. The following membership functions are defined based on the speed of athletes: [7]
   Low={0+0.1+0.3}, Medium={0.5+0.57+0.65}, High={0.8+0.9+1.0}
   Find the following:
   (a) R~=Low x Medium
   (b) S~=Medium x High
   (c) T~=R o S using max-min composition

9. (a) List the stage involved in Back Propagation Algorithm. [5]

   (b) State the training algorithm for multiple output classes in Perceptron. [5]

   (c) What is the role of Widrow-Hoff rule in Adaptive Linear neuron? Give appropriate equations. [5]

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10. (a) Discuss Ant Colony Optimization (ACO) in details. [5]

(b) Discuss Particle Swarm Optimization (PSO) in details. [5]

(c) Discuss the concept of evolutionary search methods. [5]

11. (a) Draw the flowchart of Hebb training algorithm. [5]

(b) Design a Hebb network to implement NOR function using with bipolar inputs and targets. [5]

(c) Find the weights required to perform the following classifications using perceptron network: [5]
The vectors (1, 1, -1, -1) and (1,-1. 1, -1) are belonging to a class having target value 1. The vectors (-1, -1, -1, 1) and (-1, -1, 1, 1) are belonging to a class having target value -1. Assume learning rate 1 and initial weights as 0.

*** END OF PAPER ***

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